RAUNAK RAJ

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PROFESSIONAL SUMMARY

Cybersecurity analyst with production-grade deployment experience across digital forensics, SOC operations, and offensive security engineering — including a government engagement at IFSO, Delhi Police (Ministry of Home Affairs) where I delivered forensic tooling used in active law enforcement prosecutions. Built NebulaX, an AI-augmented full-cycle penetration testing platform, and WLFS, a volatile memory forensics suite deployed in production at a national cybercrime unit — both engineered from scratch. Fluent across MITRE ATT&CK, NIST CSF, and Cyber Kill Chain in both red and blue team contexts. ISC2 Certified. Open to enterprise security engineering, SOC analyst, or red team roles where security rigour is a commercial differentiator, not a compliance checkbox.

TECHNICAL SKILLS

Offensive Security:	Penetration Testing, VAPT, Metasploit, Burp Suite, OWASP ZAP, Shodan, theHarvester, Sublist3r, SQL Injection, XSS, Privilege Escalation, Lateral Movement, Active Directory Enumeration, API Security Testing, LSB Steganography
Forensics & IR:	Volatile Memory Acquisition (WinPmem), Autopsy, Volatility 2/3, MFT Parsing, Registry & Browser Artifact Analysis, Malware Triage, IOC Identification, Timeline Correlation, MD5/SHA-256 Integrity Verification, Chain of Custody
Security Ops:	SIEM (Splunk), Log Correlation, Threat Hunting, Windows Event Log Analysis, IDS/IPS (Snort), Packet Analysis (Wireshark), EDR Concepts, Alert Triage, Incident Escalation & Containment, Playbook-Aligned Response
Threat Intelligence:	VirusTotal, AbuseIPDB, OSINT (Shodan, theHarvester, Sublist3r), IOC Enrichment, CVSS v3.1 Scoring, Threat Actor Profiling
Frameworks:	MITRE ATT&CK, OWASP Top 10, Cyber Kill Chain, NIST CSF, CVSS v3.1, Zero Trust Architecture, GRC Fundamentals, Cloud Security Concepts (AWS/Azure)
Networking:	TCP/IP, Firewalls, Stateful Packet Inspection, IDS/IPS, VLAN Segmentation, DHCP, HTTP/HTTPS, Network Mapping, East-West Traffic Analysis
Development:	Python, Bash, JavaScript, Node.js, React, MongoDB, Express.js, REST APIs, JWT, Puppeteer, PySide6
Documentation:	Technical Report Writing, Forensic Evidence Documentation, Prosecution-Ready Case Files, Stakeholder Incident Summaries

WORK EXPERIENCE

Cybersecurity & Digital Forensic Intern | *I4C NCFL-IFSO, Cyber Special Unit — Ministry of Home Affairs* | Delhi — Dec 2023 – Apr 2024

- Engineered the WinLive Forensic Suite (WLFS) for IFSO, Delhi Police — consolidating RAM acquisition, browser artifact extraction, and registry hive collection from a **4–6 tool manual workflow into a single Python pipeline**, reducing live evidence preparation time by **∼70%** and deploying across **5+ active cybercrime investigations** within the first month of internship.
- Automated volatile memory capture on **10+ live suspect systems** using the WinPmem kernel driver, producing Volatility 2/3-compatible forensic images in under **8 minutes per acquisition** — eliminating all manual kernel-level collection steps and maintaining chain-of-custody integrity for court-admissible evidence packages.
- Identified **30+ indicators of compromise (IOCs)** across active cybercrime investigations through live-system forensic triage, delivering MD5/SHA-256-verified timeline reports used directly in Delhi Police case documentation and prosecution support files.
- Recognised with a formal **Letter of Appreciation from NCFL-IFSO, Delhi Police (Ministry of Home Affairs)** for direct technical contribution to live operational investigations — one of select interns to receive commendation for production-level forensic impact.

Security Administration Associate — L1 | *Infotact Solution* | Bengaluru (Remote) — Jul 2024 – Oct 2024

- Deployed and tuned a Snort-based NIDS/IPS protecting a **100+ endpoint enterprise network**, detecting and flagging **200+ intrusion attempts** over the 4-month engagement — reducing mean analyst escalation time by **∼40%** through threat-severity triage and false-positive filtering aligned to the MITRE ATT&CK kill chain.
- Engineered a custom Python stateful firewall with granular ACL rules across **6 network segments**, enforcing east-west traffic controls that blocked all unauthorised lateral movement attempts across monitored VLANs — achieving **zero confirmed breaches** throughout the deployment period.
- Triaged **30+ daily Splunk SIEM alerts** through log correlation and playbook-aligned categorisation, escalating validated P1/P2 threats to senior analysts within a consistent **sub-15-minute MTTE** — processing **2,500+ alerts** across the engagement with zero missed critical incidents.

PROJECTS

NebulaX — AI-Augmented Recon-to-Report Penetration Testing Engine — *Ongoing*

- Architected a full-cycle MERN penetration testing platform integrating OSINT, Sublist3r subdomain enumeration, Nmap network scanning, and OWASP ZAP web vulnerability detection — compressing a complete recon-to-report engagement from a **multi-day**

manual process to a single automated run under 90 minutes, covering **50+ vulnerability classes** per target.

- Engineered a ReAct-loop AI Decision Engine that autonomously sequences attack steps across a **12-phase pentest kill chain** based on live scan output — eliminating manual analyst pivot decisions between phases and replicating senior analyst decision logic without human intervention.
- Integrated VirusTotal, AbuseIPDB, and NIST NVD APIs for real-time threat enrichment with CVSS v3.1 scoring; Puppeteer-automated PDF generation produces a structured **client-ready pentest report in under 60 seconds** post-scan — reducing report production time from **hours to under a minute**.

WinLive Forensic Suite (WLFS) *Production — IFSO, Delhi Police*

- Architected a multi-threaded PySide6 forensic acquisition tool with automated evidence collection for System Inventory, Browser Forensics, and Windows Registry artifacts — deployed in production at a Ministry of Home Affairs cybercrime unit and used in **5+ live law enforcement investigations**.
- Integrated WinPmem kernel driver for RAW volatile memory capture with full Volatility 2/3 compatibility, enabling deep offline analysis of running process trees, injected code artifacts, and live network connection states from acquired **forensic images averaging 8–32 GB** per suspect system.
- Built a Timeline Correlation Engine and MFT parsing module that reconstructs **100,000+ filesystem events** per acquisition into chronological investigation sequences, with MD5/SHA-256 verification producing documentation that meets Delhi Police court-admissibility standards.

LSB Steganography Offensive Security Tool *Security Research*

- Demonstrated LSB steganography-based payload exfiltration paired with live system enumeration and network mapping to expose detection gaps in signature-based AV environments — **confirmed bypass viability against ClamAV and Windows Defender** signature scanning with **0% detection rate** on carrier files under 5 MB.
- Developed a payload extraction module that recovers hidden data from carrier files while preserving metadata integrity, validating covert channel techniques used by advanced threat actors to operate below conventional IDS/signature-detection thresholds.

EDUCATION

Bachelor of Computer Applications (BCA) — Cyber Security & Forensics | *BBD University* | Lucknow, UP 2023 – 2026

- CGPA: **8.73 / 10** (Top 10% of department)
- **Discipline Head, Lakshya** (Student Technical Body) — directed cross-departmental security awareness and technical programmes across the university's computing faculty, coordinating **3+ annual events**

CERTIFICATIONS & TRAINING

- **Certified in Cybersecurity (CC)** — ISC2
- **SOC in Practice** — IBM
- Cybersecurity for Businesses, Fundamental Edition
- **Fundamentals of Artificial Intelligence** — IBM

Hands-on Training: SOC & VAPT Program — SIEM alert triage, threat hunting, incident response, OWASP Top 10 exploitation (SQLi, XSS, privilege escalation, lateral movement), and MITRE ATT&CK / NIST CSF application in live red-vs-blue attack simulations.

ACHIEVEMENTS

- **Letter of Appreciation — NCFL-IFSO, Delhi Police / Ministry of Home Affairs** — awarded for direct technical contribution to live operational cybercrime investigations
- **Best Excellence Award** — Digipodium, for outstanding contribution in cybersecurity
- **Student Head** — 12th Grade, led institutional operations and cross-team coordination